



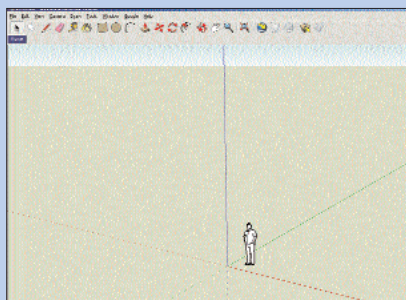
Building Homes In Google SketchUp

Free 3D design tools have never been this easy to use

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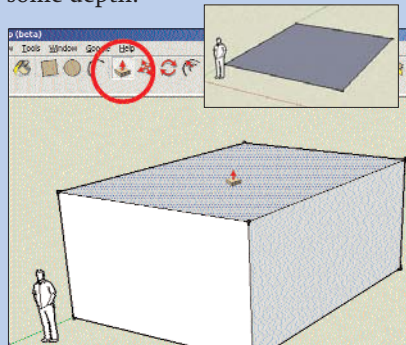
They did it again, minus the “oops.” Google’s SketchUp will go down as yet another of Google’s victories, this time in the world of 3D modelling. You will find SketchUp on the June 2006 CD. Here’s the dirt you need to get cracking at creating your own monument.

STEP 1 Getting Around Quickly When you start SketchUp, all you will see is a 2D human figure. To move around the centre of the scene (called *orbiting*), hold down the middle mouse button and drag the mouse. To move from side to side (called *panning*), hold down [Shift] as well. Alternatively, you could press [O] to select the Orbit Tool and use the left mouse click, and hold down [Shift] when you want to pan.



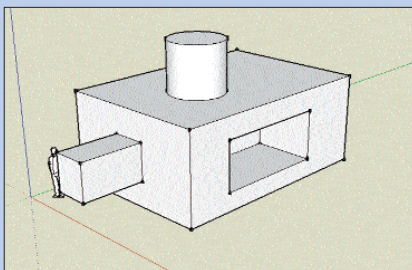
The default start-up screen

STEP 2 Start Building! To lay the first block (which you will then manipulate to turn into a house), first select the Rectangle Tool (shortcut: [R]) and draw the rectangle as shown in the screenshot alongside. Then use the Push Tool (shortcut: [P]) to give it some depth.



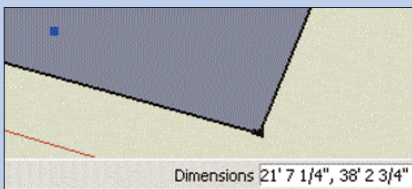
draw the rectangle, then push!

STEP 3 More Complex Building You can now draw rectangles (or circles (shortcut: [C]), for that matter) on the surface of the box you just made. To be absolutely sure you’re drawing on the surface, just hover the cursor where you’re drawing, until a tool tip tells you where you’re drawing. And once you draw these rectangles, you can use the Push Tool again to create protrusions and dents.



Use the Rectangle and Push Tools to create more complex structures

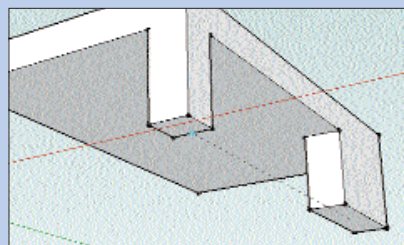
STEP 4 Real-World Dimensions SketchUp is meant for building houses, right? So where does the ever-important “accurate to scale” part come in? Hidden in the bottom-right corner, you will see a little box that contains the dimensions of the figure you’re drawing. If you’ve just drawn a new rectangle, just key in the dimensions you want—if you wanted 12 feet by 15 feet, you’d type “12”, “15”. You can also use other units—“m” for metres, “in” for inches and so on. This applies no matter which tool you’re using.



The little dimension box that nobody sees

STEP 5 All Are Equal If you’re modelling something like a platform, chair or table, you will obviously need to have legs of equal height. One way to do this is by using dimensions as above. However, you can also use the following simple method: start with the first leg and

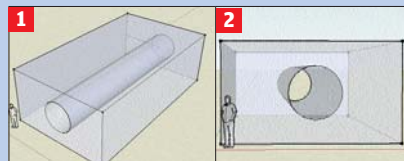
use the Push Tool to push it to where you want it. Now click on the next leg, and hover the mouse cursor at the bottom of the first. The new leg will automatically extend to that height. Lather, rinse, repeat!



Hover the mouse cursor to make the heights match

STEP 6 Cutting Holes In Objects To cut holes into your models, first draw a shape on the surface of the model. Now use the Push Tool (shortcut: [P]) and push this shape towards the surface at the other end. When the two surfaces meet, let go of the mouse button. Peek-a-boo!

To prevent unnecessary confusion, turn on the X-Ray display mode: select View > X-Ray. To make sure the surfaces intersect properly, use the method described in Step 5.



1. First draw the hole and push...
2. ...then marvel at your handiwork

STEP 7 Grouping Objects Once you’re done poking holes and making random protrusions on objects, it’s a wise idea to group them to prevent any accidental editing. Hit the spacebar to switch to the Selection Tool, and draw a selection box around the object to select all its edges and surfaces. Now right-click on it and choose “Make Component”. Give it a name, and now you can’t accidentally edit this object. To edit it, double-click on the component with the Selection Tool—you can now edit it while the rest of the objects in your scene fade out. ■

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